Example 3.70
Suppose that a boat travels $10\ km$ from the port towards east and then turns $60°$ to its left. If the boat travels further $8\ km$, how far from the port is the boat?

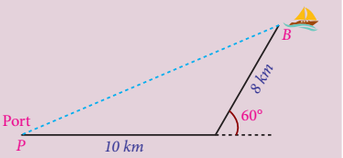


Figure 3.25

Solution:
Let BP be the required distance.
By using the cosine formula, we have,

$$BP^2 = 10^2 + 8^2 - 2 \times \times 10 \times 8 \times \cos 120° = 244km \Rightarrow BP = 2\sqrt{61}\ km$$

Example 3.71
Suppose two radar stations located $100km$ apart, each detect a fighter aircraft between them. The angle of elevation measured by the first station is $30°$, whereas the angle of elevation measured by the second station is $45°$. Find the altitude of the aircraft at that instant.

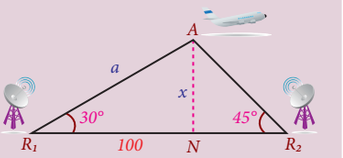


Figure 3.26

Solution:
Let R_1 and R_2 be two radar stations and A be the position of fighter aircraft at the time of detection.
Let x be the required altitude of the aircraft.
Draw $\perp AN$ from A to R_1R_2 meeting at N.

$$\angle A = 180° - (30° + 45°) = 105°$$

$$\text{Thus, } \frac{a}{\sin 45°} = \frac{100}{\sin 105°} \Rightarrow a = \frac{100}{\frac{\sqrt{3}+1}{2\sqrt{2}}} \times \frac{1}{\sqrt{2}} = \frac{200\left(\sqrt{3}-1\right)}{2} = 100 \times \left(\sqrt{3}-1\right) km$$

$$\text{Now, } \sin 30° = \frac{x}{a} \Rightarrow x = 50 \times \left(\sqrt{3}-1\right) km.$$

Exercise - 3.10

1. Determine whether the following measurements produce one triangle, two triangles or no triangle: $\angle B = 88°, a = 23, b = 2$. Solve if solution exists.

2. If the sides of a $\triangle ABC$ are $a = 4, b = 6$ and $c = 8$, then show that $4\cos B + 3\cos C = 2$.
3. In a $\triangle ABC$, if $a = \sqrt{3} - 1, b = \sqrt{3} + 1$ and $C = 60°$, find the other side and other two angles.
4. In any $\triangle ABC$, prove that the area $\triangle = \dfrac{b^2 + c^2 - a^2}{4\cot A}$.

5. In a $\triangle ABC$, if $a = 12\ cm$, $b = 8\ cm$ and $C = 30°$, then show that its area is $24\ sq.cm$.
6. In a $\triangle ABC$, if $a = 18\ cm$, $b = 24\ cm$ and $c = 30\ cm$, then show that its area is $216\ sq.cm$.
7. Two soldiers A and B in two different underground bunkers on a straight road, spot an intruder at the top of a hill. The angle of elevation of the intruder from A and B to the ground level in the eastern direction are $30°$ and $45°$ respectively. If A and B stand $5\ km$ apart, find the distance of the intruder from B.
8. A researcher wants to determine the width of a pond from east to west, which cannot be done by actual measurement. From a point P, he finds the distance to the eastern-most point of the pond to